VR Part II: Raising VR onto Zero-Trust Logic.

Witnessed identity as earned truth, choice sequences as the lazy register, and a survival ledger for the operational corpus

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2026 — Version 1.0.0

Abstract

The VR cycle built an operational mathematics — arithmetic, numbers, sets, forms, topology, a continuum on Brouwer's path — and only afterwards wrote out the logic it had been standing on: ZTL, Zero-Trust Logic, a two-valued logic over marked (unverified) inputs generated by the single principle *truth is never granted on credit* (Zenodo, concept DOI 10.5281/zenodo.21318981). This preprint carries out the programme "raise VR onto ZTL" and verifies, rather than declares, the thesis that *VR always stood on ZTL.* Three steps, every claim either MEASURED (machine enumeration, reproducible by the test stands of the ZTL repository) or kernel-checked in Lean 4 with the axiom footprint printed per object. (a) *Witnessed identity is a ZTL atom discipline*: the operational set universe `VRCycle.SetsOp` carries identity as a bisimulation witness; we package verdicts *with* their certificates (`EqVerdict`, `MemVerdict`), show that the ZTL inference rules alive on identity atoms are exactly witness constructors, that identity on finite operational sets and on the von Neumann register is *totally earnable*, that groundedness of a set is orthogonal to earnability of its identity (the cycle that dooms the liar is harmless in $\Omega = \{\Omega\}$), and that a fully earned register is classical — greedy zero-trust verdicts over decided atoms neither lie nor refuse. The entire verdict layer sits on the *empty* axiom list. Step (a) also returned a correction to ZTL itself: its verdict-warranty equivalence claim proved pool-relative; the warranty is a two-grade ladder (sound — never lies; hereditary — never revoked), published same-day as ZTL v1.1 (DOI 10.5281/zenodo.21323552). (b) *Choice sequences are the lazy register*: for a growing binary sequence, what a stage may assert is what the prefix forces over admitted continuations; we measure and then kernel-check that the lawless stage court coincides with ZTL's global supervaluation *totally* (a law is knowledge: it narrows the worlds and strictly strengthens the stage), that Kripke persistence — refused as a free axiom by the greedy register — is native to the lazy one, that warranted greedy verdicts are exactly the Brouwer-assertable ones, and that the fallen law of identity $p \to p$ is *redeemed* by the stage court: a law of logic, not of data. (c) *The survival ledger*: a proof survives the move onto ZTL iff it stands below the classical floor — ZTL's fallen laws are laws of free truth, their Lean fingerprint is `Classical.choice`, and a constructive proof from earned premises is a chain of ZTL-alive rules. The cycle's four-tier axiom ledger was therefore the ZTL-survival audit all along; sweeping 405 live-audited objects plus flagship anchors shows that everything VR calls operational moves, and what stays is exactly what the cycle had already flagged as classical by design, by substrate, or by borrowed plumbing. No operational theorem died in the move.

Keywords: zero-trust logic, three-valued logic, witnessed bisimulation, operational set theory, choice sequences, supervaluation, Kripke persistence, intuitionism, constructive mathematics, axiom audit, Lean 4, machine-checked proofs.

1 Introduction

Brouwer built a mathematics first; Heyting wrote out its logic afterwards. The VR cycle repeated the pattern at its own scale: ten works of operational mathematics — an axiom-free reconstruction of arithmetic, operational number superstructures, two set-theoretic registers, a two-register apparatus of forms, formal topology, a continuum on Brouwer's path — were built under a uniform discipline (operations before objects, witnesses before truths, an axiom audit per object) before the logic of that discipline was written down. The logic, when it was finally written out and machine-checked, is ZTL — *Zero-Trust Logic* [1]: exactly two truth values (verdicts are always classical), a third symbol Z that is a *mark on an unverified input* rather than a truth value, and one generating principle for all connectives — *a connective returns T only if T is forced under every classical reading of the unverified.* Truth is never granted on credit.

The present work is Part II of that story: the programme "raise VR onto ZTL," stated by the curator as three steps and a thesis. The thesis — *VR always stood on ZTL* — is attractive and dangerous in equal measure: attractive because the two disciplines visibly rhyme (witnessed identity looks like earned truth; apartness earned at a finite stage looks like a verdict with a certificate), dangerous because rhymes are cheap. The VR cycle's standing rule is that such claims are *verified, not postulated.* Accordingly, everything below carries one of two tags: MEASURED — established by exhaustive machine enumeration, reproducible by the ZTL repository's regression stands (E20–E22) — or a Lean 4 theorem, with `#print axioms` output quoted per object.

The three steps, and one find that travelled in the opposite direction:

- **Step (a)** (§4): the witnessed identity of the operational set universe *is* a ZTL atom discipline. Verdicts are packaged with their certificates; soundness is the shape of the type, not a theorem about it; the alive inference rules are witness constructors; finite and von Neumann identity is totally earnable; a fully earned register is classical. The whole layer sits on the *empty* axiom list.
- **The return find** (§4.2): the identity atoms falsified a claim of ZTL v1.0 — the warranty of a verdict is not one bit but a two-grade ladder. The correction was published the same day as ZTL v1.1. Part II thus begins by demonstrating the direction of authority: the measured corpus corrects the logic, not the other way round.
- **Step (b)** (§5): Brouwer's two objects — the sequence-in-growth and the assertion-at-a-stage — are ZTL's two registers. The lawless stage court coincides with global supervaluation totally; persistence is native to the lazy register; $p \to p$ is redeemed at the stage.
- **Step (c)** (§6): the survival criterion and the ledger. The cycle's own four-tier axiom audit turns out to be the ZTL-survival audit; the operational corpus moves wholesale.

Before the steps, §3 discharges a debt any such programme owes: ZTL's slogan could be Brouwer's, so one must show the logic is not intuitionism rediscovered. The answer is measured: the two are *incomparable* as law-sets, agree exactly on a battery of premised rules, and part ways on every structural signature that defines intuitionism.

Code and reproducibility. The ZTL side lives in the ZTL repository (test stands `zipc.py`, `zopsets.py`, `zchoice.py` — expeditions E20–E22 — inside a 30-stand regression, plus ten Lean modules on the empty axiom list). The VR side lives in the VRCycle repository: the Lean module VRCycle.SetsZTL (`Kernel`, `Atoms`, `Stages`, `Survival`), the survival ledger `ZTL_SURVIVAL.md`, and blueprint chapter 14. The ZTL value kernel is *vendored* into VRCycle with a commit pin; the dependency is one-way — ZTL never depends on VR.

2 The two sides of the stitch

ZTL in one paragraph. Values T, F; input mark Z. For a classical connective f, its ZTL lift returns T iff f returns true under *every* classical reading of the Z-arguments (each occurrence read independently); otherwise F. Compound formulas are always classical: Z lives only on atoms. The measured price list of v1.0: 12 classical laws survive, 14 fall, and every fallen law is a law of "free truth" — it mints a verdict for an unverified atom ($p \to p$, $q \to (p \to q)$, excluded middle, double-negation elimination, both De Morgan laws as identities, idempotence...). The discovered split: of 14 classical *premised rules*, 12 survive — inference transports earned truth; laws mint it. Verification is an act ($\mathsf{Z} \to \mathsf{T}/\mathsf{F}$), and a verdict is a pair (value, warranty); after the Part-II correction (§4.2) the warranty is a two-grade ladder. Everything is machine-checked in Lean 4 with an *empty* axiom list [1].

VR in one paragraph. The operational register of VR's set theory, VRCycle.SetsOp [2], carries a set as a *revealing functionality* (a pointed graph) and identity as a *witnessed bisimulation* — a relation carried with its witness, never collapsed to a quotient (the collapse is what pulls choice). Foundation is a predicate, not an axiom: the ZFC-like and ZFA-like universes are fragments of one domain. The continuum [4] is built on Brouwer's path: branches of the binary spread with their finite performed segments. The cycle's uniform discipline is the axiom audit: every public object prints its axioms, and the corpus is stratified into four tiers, from the empty list to the classical tier with `Classical.choice`.

The stitch, in one sentence. An identity atom $x \approx y$ of the operational universe *is* a ZTL atom: earned T carries the bisimulation, earned F carries the refutation (apartness at a finite separation stage), and the unexamined atom is worth exactly the input mark Z.

3 Not intuitionism rediscovered (MEASURED)

ZTL's slogan — no truth on credit — could be read as Brouwer's, and VR's continuum walks Brouwer's road; so Part II owes an explicit delta. It is measured by expedition E20: a calibrated decision procedure for intuitionistic propositional logic (Dyckhoff's contraction-free G4ip [8]; "IPC refuses" below means *non-theorem*, not "search failed") against ZTL's native entailment, on a 27-law canonical battery and the 14-rule premised battery.

- **Law-sets are incomparable** inside classical logic, both directions witnessed. IPC-valid but ZTL-refused: 6 witnesses, including $p \to p$ and $q \to (p \to q)$. ZTL-valid but IPC-refused: the weak law of excluded middle $\neg p \vee \neg\neg p$ — Jankov's axiom, the axiom of the intermediate logic KC [9]. Since every intermediate logic proves $p \to p$, ZTL is *not* an intermediate logic: it sits beside IPC, not above it.
- **Premised rules coincide 14/14**, including both casualties ($\neg\neg$-elimination; tautology-in-conclusion). The diagnoses differ — no construction of p versus a laundered mark — but the casualty list is identical: both systems refuse inference moves that mint a certificate out of nothing. (As full consequence relations they differ already at empty premises: $p \to p$.)

- **Every structural signature separates.** ZTL is a finite matrix; IPC provably has none (Gödel 1932 [6]). IPC has the disjunction property; ZTL loses it (measured witness and census: its $\vee$ is supervaluational, not BHK). ZTL's double negation is verdict-transparent ($\models A \iff \models \neg\neg A$, checked totally on a 786-formula pool); IPC's is load-bearing (Glivenko [7], verified 27/27 by the prover as a workout). An unproved sentence in intuitionism is an open potentiality; a ZTL refusal is a terminating answer with a passport.

Same slogan, different creditors: intuitionism refuses truth without a *construction*; ZTL refuses it without *certified input*. Proof-debt versus data-debt. This is the licence for Part II: raising VR onto ZTL is a move onto a third foundation posture, not a relabelling of the second.

4 Step (a): witnessed identity as earned truth

4.1 The measured half (E21)

Finite operational sets are modelled exactly as `OpSet` builds them — pointed reveal-graphs — and identity atoms are judged by a bisimulation earner. MEASURED on a zoo including the von Neumann naturals, duplicated representations, and the Quine atom $\Omega = \{\Omega\}$:

- **Identity is totally earnable**: every atom decides — T with a checkable witness (the relation is re-verified independently of the search), F with a finite separation stage, in the apartness discipline of ZTL's stream expedition. On finite carriers Z is an input state, not a fate. (Contrast streams, §5: there equality is Z-permanent — finiteness is what buys totality, not the logic.)
- **Deduplication is earned**: $\{\emptyset, \emptyset\} \approx \{\emptyset\}$ with an explicit witness. ZTL's set expedition had measured $\{\mathsf{Z}, \mathsf{Z}\} \neq \{\mathsf{Z}\}$ — merging *unverified* members is not earned; together the two halves close one story: deduplication happens exactly when identity is witnessed.
- **Alive rules are witness constructors**: reflexivity is the diagonal, symmetry the converse, transitivity the composition, congruence the union — 44 constructed witnesses on the zoo, every one re-checked. The ZTL rules alive on $\approx$-atoms are not table luck.
- **Cardinality is an interval that verification narrows monotonically**: $|\{x, y, z\}| \in [1, 3] \supseteq [1, 2] \supseteq [2, 2]$ along truthful verification of the identity atoms, with the last atom earned *free* by partition consistency — witness composition working by itself.
- **Groundedness $\perp$ earnability**: $\Omega \approx \Omega_2$ (the two-node cycle) is earned T while `IsGrounded`(Ω) is false. The same cycle that dooms a sentence (the liar: permanent quarantine, a least-fixed-point leftover) is harmless in a set (identity is greatest-fixed-point-shaped). Anti-Foundation-as-theorem is the set-side mirror of quarantine-as-theorem: two registers, one architecture.

4.2 The return find: the warranty is a ladder

The identity atoms then did something unplanned: they falsified a claim of ZTL v1.0. That version asserted an equivalence — "a verdict is stable under the global supervaluation iff it is invariant under every sequence of verifications" — measured totally on its ten-formula pool, with a monotonicity corollary ("a stable verdict is never revoked"). On the 3303-formula pool of E21, 96 supervaluation-stable T-verdicts *die* under truthful verification. The separating shape is **or(ladder, gap)**:

$$\neg\neg p \;\vee\; (q \vee \neg q)$$

— greedy T via the $\neg\neg$ ladder, insured by a gap that is true in *all* completions yet greedy-F; verifying $p := \mathsf{F}$ kicks the ladder away before the gap closes. A simpler cell: $(\neg p) \to (q \to q)$, where the gap is the fallen law of identity itself.

Finding 4.1 (The two-grade warranty)*.* Sound (supervaluational: every completion agrees with the verdict) buys *never lies* — measured: of 9192 sound-warranted T-verdicts, none is false of the facts. Hereditary (invariant under every partial refinement) buys *never spoils* — measured: 0 deaths, totally. Hereditary implies sound; the converse is false (the grades separate).

The correction was cross-checked with the original section's own instruments, adopted into the ZTL code the same day (the studio now speaks in grades: "hereditary T — build on it"; "sound T — never a lie, may stall"), and published as ZTL v1.1 [1]. Part II's first contribution is thus a correction flowing *from* the mathematics *into* the logic — the direction of authority the whole cycle is built on.

4.3 The kernel-checked half (SetsZTL.Atoms)

The Lean module packages a verdict *with* its certificate:

Definition 4.2 (`EqVerdict`)**.** *For operational sets x, y: `earnedT` holds a bisimulation witness $x \approx y$; `earnedF` holds a refutation; `unexamined` is the zero-trust start. `val` sends the three to T, F, Z. Soundness is not a theorem about the type but its shape: `sound_T` and `sound_F`* extract *the certificate from the value.*

Theorem 4.3 (constructors; vn register; orthogonality)**.** *`refl`, `symm` (verdict-preserving), and `composeT` (composition of carried witnesses) are certificate constructors for the alive rules. On the operational naturals the identity verdict `vnVerdict n m` is never Z and equals T exactly when $n = m$. The Quine atom earns its identity, and its apartness from $\emptyset$, while ungrounded.*

Theorem 4.4 (The fully earned register is classical)**.** *If every atom of a formula carries a decided, truthful verdict, the greedy ZTL value of every formula over the six connectives is decided and truthful: T exactly on the classically true formulas, Z nowhere. Over verified identity atoms the zero-trust connectives neither lie nor refuse.*

Axiom profile (printed per object). The entire verdict layer — including the classicality theorem — reports *"does not depend on any axioms"*: no choice, no quotients, not even propositional extensionality. The vn-register theorems initially inherited `propext` from the host module's proof style (`rw` by *iff*-lemmas); a tier pass replaced those rewrites by explicit *iff*-combinators, after which *all 39 audited objects of SetsOp — including `vn_inj` and the non-enumerability of the operational universe — and all of SetsZTL sit on the empty axiom list.* The only remaining non-empty profile in the wing is the describable-register encoding (borrowed pairing plumbing from the mathlib library — constructively true upstream facts proved there with classical instruments; documented, removable at a known price).

5 Step (b): choice sequences as the lazy register

Brouwer's continuum distinguishes the sequence-in-growth from what is assertable at a stage. VR's continuum took that road; ZTL, it turns out, had already priced it: the distinction *is* the two-register split. MEASURED by expedition E22 (pool of 3303 formulas over three position atoms; all 15 prefixes of a binary horizon; the stage court judging by forcing over admitted continuations), then kernel-checked in SetsZTL.Stages against VR's own `Branch` of the binary spread.

- **The ladder of judges runs through the warranty.** The naive inclusion "greedy $\mathsf{T} \Rightarrow$ stage-forced T" is false: 2027 of 27905 raw greedy T-verdicts assert what no prefix forces (witness: $\neg\neg a_0$ at the root — the E12 cell read temporally, a verdict about a future the data does not underwrite). The true chain has 0 violations: *warranted* (sound-grade) greedy $\mathsf{T} \Rightarrow$ stage-forced $\mathsf{T} \Rightarrow$ true on every admitted continuation. Zero trust speaks assertably about a growing sequence exactly when it carries its warranty — the ladder of §4.2 vindicated from Brouwer's side within a week of its discovery.

- **Lawless = global supervaluation, totally** (49545 formula-prefix pairs, 0 mismatches): for a lawless sequence the stage court *is* ZTL's global supervaluation — zero trust is assertability against a maximally ignorant future. **A law is knowledge**: under "exactly one 1" the stage court forces a disjunction at stage 0 that the supervaluation refuses (1968 such cells) — Brouwer's lawless/lawful divide lands on ZTL's modal frame as: *which world-set the global $\Box$ ranges over.*

- **Heredity is native to the lazy register**: stage verdicts are never revoked by a revealed bit (0 revocations, against 1371 greedy deaths). Kripke persistence — which the greedy register refuses as a free axiom and sells back as the warranty bit — is a theorem of the stage court (`forces_mono`, kernel-checked).

- $p \to p$ **is redeemed.** At the root every branch satisfies the identity law while the greedy court abstains (`zimp Z Z = F`): a law of logic, not of data — the stage court asserts it, the data court waits. Both cells on the empty axiom list (`identity_forced_at_root`, `identity_greedy_abstains`).

- **Streams reproduce the constructive asymmetry**: equality of branches is never forced at a finite stage (two explicit continuations diverge below any node — `eq_never_forced`), apartness is earned by one disagreeing reveal and, being an existential fact, persists (`apart_earned`). Measured coupling: eq $= \mathsf{F} \iff$ apart $= \mathsf{T}$, yet apartness is *not* the ZTL-negation of equality even at the level of marks ($\neg\mathsf{Z} = \mathsf{F} \neq \mathsf{Z}$; 227 cells) — the constructive primitivity of # falls out of the zero-trust tables.

Axiom profile. The kernel cells are on the empty list; the `Branch`-layer theorems (`stage_eq_super`, `forces_mono`, the stream pair) inherit [`propext`(, `Quot.sound`)] from the mathlib list lemmas — the Continuum wing's own choice-free tier, with `Classical.choice` nowhere (one `simpa` was evicted for pulling it). Measured lemma-by-lemma: the residue is upstream proof style, not logic; removable at a documented price.

6 Step (c): the survival ledger

What does it mean for an operational theorem to *survive* the move onto ZTL? A naive answer — audit the proofs for uses of the fallen laws — is a category error: Lean proofs live in Lean's calculus and never manipulate Z. The criterion adopted (the curator's decision, after the alternatives were weighed) is:

Definition 6.1 (Survival criterion C2)**.** *A proof survives the move onto ZTL iff it stands below the classical floor: no `Classical.choice` in its measured axiom footprint.*

The bridge that justifies it, each link measured: ZTL's fallen laws are exactly the laws of *free truth*, and their Lean fingerprint is classical case analysis — applying excluded middle to a proposition is declaring every atom pre-verified. At the level of premised rules ZTL keeps what constructive inference keeps (§3: 14/14); alive rules are witness constructors (§4); so a constructive proof from earned premises *is* a chain of ZTL-alive rules, delivering the conclusion's certificate. VR's operational theorems are premised by construction — operationality is earned inputs — so they are rules, not laws: the surviving category.

Corollary 6.2 (The thesis, made literal)**.** *The four-tier axiom ledger the VR cycle has kept since its first work is the ZTL-survival audit. "VR always stood on ZTL" is, under C2, a bookkeeping theorem rather than a slogan.*

The ledger (MEASURED). 405 public objects live-audited by re-elaborating the whole project, plus flagship probes of the wings whose audits live in external ledgers. Moves wholesale: the VR formal system (the flagship VR $\leftrightarrow$ PA equivalence is *axiom-free*), VR-Forms (45/45 clean), the operational set universe (39/41; the two are the describable-register plumbing), Part II itself (38/38), the operational continuum (88/92), formal topology (its complete ledger reports zero choice; the binary Tychonoff re-probed clean), the integer line, and — notably — the ZF-closure facts of the *classical* register, which are themselves choice-free over their quotient carrier. Stays on the classical shore, exactly as the cycle had already flagged: VR-Audit (0/17 — classical by design: choice is its subject matter), VR-Sets-ZFA (3/14 — the coinductive substrate that motivated `OpSet` in the first place), the $\mathbb{Q}/\mathbb{R}/\mathbb{C}$ correspondence theorems (substrate leak of the host library's carriers; the choice-free number line already lives in the Continuum), the Sperner–Brouwer fixed point (the real-number substrate), and assorted mathlib-carrier instances in Algebra/Apparatus. *No operational theorem died in the move; the fallen laws of ZTL were never load-bearing in the operational corpus.*

Spot-checks (SetsZTL.Survival, all on the empty axiom list). Three flagships exhibited in the moved form. *Strong extensionality*: a law in the classical register, a *rule* in the operational one — `extVerdict` constructs the identity certificate from the member-matching data. *Anti-Foundation*: `afaVerdict` — two decorations of one graph carry earned identity at every vertex. *Membership*: `MemVerdict`, the membership twin of `EqVerdict` — the vn membership register is total and classical, and earned $\in$ transports along earned $\approx$ (`memCongrVerdict`, the rule form of congruence). The continuum spot-check, apartness earned at a stage, is §5.

The reading discipline. The audit bounds the *given* proof, never the theorem: a classical tier means *this proof does not move*, not *the theorem is false on ZTL* — the witness method is an auditor, not a separator. And the audit is blind to impredicativity (the cycle's power-set finding); ZTL says nothing about impredicativity either — a shared boundary, recorded.

7 What Part II does not claim

- The rule-agreement with IPC and the law-incomparability witnesses are battery-relative in their counts (the incomparability itself is existential, hence robust).
- The warranty ladder's Lean twin is deliberately *not* formalised here: it belongs to the ZTL corpus, where it is measured; VR consumes it.
- The stage court's equivalence with supervaluation is kernel-checked for horizon-read properties over the binary spread; richer laws enter only as world-set restrictions, as in the measured example.

- Criterion C2 audits proofs, not theorems (above); a semantic criterion — a ZTL-valued model for VR statements, truth in a model instead of proof audit — is stronger, was weighed, and is deferred as a separate work.
- Wings whose audits live in external ledgers are anchored by flagship probes, not by a full live sweep.

8 Conclusion

Part I of the VR story built an operational mathematics and paid for every theorem in the currency of an axiom audit. Part II shows the currency was a logic. Witnessed identity is earned truth (and packaging verdicts with their certificates costs *zero* axioms); Brouwer's growth/stage distinction is the lazy/greedy register split (and the lawless stage court is, totally, the global supervaluation); the corpus's four-tier ledger is the survival audit, under which everything operational moves and everything flagged classical stays, for the reasons already on record. Along the way the mathematics corrected the logic once — the warranty ladder, published same-day as ZTL v1.1 — and the logic returned the favour by explaining the corpus's oldest discipline to itself.

Brouwer built a mathematics, and Heyting wrote out its logic. VR was built first, and ZTL is its logic written out — and the delta measured in §3 says the second pair is not a photocopy of the first.

Acknowledgement of AI assistance

This preprint, its companion Lean 4 formalisation (VRCycle.SetsZTL), the survival ledger, and the measured expeditions E20–E22 in the ZTL repository were prepared with the assistance of Claude (Anthropic), in the Variant A architecture used across the VR cycle: a human curator directing a model that served as architect and implementer. All mathematical content, definitions, theorems, fork decisions and foundational positions are due to the author, Vitaly Reznik. This work was developed with Claude Fable 5. The reliability of the results does not depend on trusting the AI: every MEASURED claim is reproducible by the repositories' regressions, and every Lean claim is checked by the Lean 4 kernel with the axiom footprint printed per object.

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